Chapter 7

(Follows from [1] Anton, …)

Chapter 7.2 Integration by Parts

$$\int f(x)g(x)dx = f(x)\int g(x)dx - \int f'(x)[\int g(x)dx]\,dx$$

$$G(x) := \int g(x)dx$$

$$\int f(x)g(x)dx = f(x)G(x) - \int f'(x)\,G(x)\,dx$$

$$u = f(x), \quad du = f'(x)\,dx$$
$$v = G(x), \quad dv = G'(x)\,dx = g(x)\,dx$$

This yields the following alternative form for (2):

$$\boxed{\int u\,dv = uv - \int v\,du}$$

Example 1

Use integration by parts to evaluate $\int x\cos x\,dx$.

$$u = x \quad \text{and} \quad dv = \cos x\,dx$$

$$du = dx \quad \text{and} \quad v = \int dv = \int \cos x\,dx = \sin x$$

$$\int \underbrace{x}_{u}\,\underbrace{\cos x\,dx}_{dv} = \underbrace{x}_{u}\,\underbrace{\sin x}_{v} - \int \underbrace{\sin x}_{v}\,\underbrace{dx}_{du}$$
$$= x\sin x - (-\cos x) + C = x\sin x + \cos x + C$$

LIATE

Logarithmic, Inverse trigonometric, Algebraic, Trigonometric, Exponential

Example 2 Evaluate $\int xe^x dx$

Solution

According to LIATE, let

$$u = x \quad \text{and} \quad dv = e^x\, dx$$

so that

$$du = dx \quad \text{and} \quad v = \int e^x\, dx = e^x$$

Thus, from (3)

$$\int xe^x\, dx = \int u\, dv = uv - \int v\, du = xe^x - \int e^x\, dx = xe^x - e^x + C$$

Example 3 Evaluate $\int \ln x\, dx$

Solution

Let, $u = \ln x$, $dv = dx$ so that $du = \frac{1}{x} dx$, $v = x$

Thus, $\int \ln x\, dx = \int udv = uv - \int vdu$

$$= (\ln x)x - \int x \frac{1}{x} dx = x \ln x - x + c$$

Repeated Integration by Parts:

Example 4 Evaluate $\int x^2 e^{-x} dx$

Solution

Let

$$u = x^2, \quad dv = e^{-x}\, dx, \quad du = 2x\, dx, \quad v = \int e^{-x}\, dx = -e^{-x}$$

$$I = \int x^2 e^{-x} dx = \int u dv = uv - \int v du$$

$$= x^2(-e^{-x}) - \int -e^{-x}(2x)\, dx$$

$$= -x^2 e^{-x} + 2\int xe^{-x}\, dx$$

$$= -x^2 e^{-x} + 2I_1, \text{ (say)} \qquad (1)$$

Now we repeat the same process to evaluate

$$I_1 = \int xe^{-x} dx \text{ as follows}$$

$$u = x, \quad dv = e^{-x}\, dx, \quad du = dx, \quad v = \int e^{-x}\, dx = -e^{-x}$$

So that, $I_1 = \int xe^{-x} dx =$

$$x(-e^{-x}) - \int -e^{-x}\, dx = -xe^{-x} + \int e^{-x}\, dx = -xe^{-x} - e^{-x} + C$$

Substituting, this I_1 into equation (1)

$$\int x^2 e^{-x}\, dx = -x^2 e^{-x} + 2\int xe^{-x}\, dx = -x^2 e^{-x} + 2(-xe^{-x} - e^{-x}) + C$$

$$= -(x^2 + 2x + 2)e^{-x} + C \quad \square$$

Example 5 Evaluate $\int e^x \cos x\, dx$

Solution

Let,

$$u = \cos x, \quad dv = e^x\, dx, \quad du = -\sin x\, dx, \quad v = \int e^x\, dx = e^x$$

Thus, $\quad I = \int e^x \cos x\, dx = \int u dv = uv - \int v du$

$$= e^x \cos x + \int e^x \sin x \, dx = e^x \cos x + I_1, \text{(say)} \qquad (1)$$

Now we repeat the process to evaluate

$$I_1 = \int e^x \sin x \, dx \text{ as follows}$$

$$u = \sin x, \quad dv = e^x \, dx, \quad du = \cos x \, dx, \quad v = \int e^x \, dx = e^x$$

$$\int e^x \sin x \, dx = \int u \, dv = uv - \int v \, du = e^x \sin x - \int e^x \cos x \, dx$$

Substituting this I_1 into (1),

$$\int e^x \cos x \, dx = e^x \cos x + e^x \sin x - \int e^x \cos x \, dx$$

$$2 \int e^x \cos x \, dx = e^x \cos x + e^x \sin x$$

$$\int e^x \cos x \, dx = \tfrac{1}{2} e^x \cos x + \tfrac{1}{2} e^x \sin x + C$$

□

Example 6 Evaluate $\int \tan^{-1} x \, dx$

Solution

Let

$$u = \tan^{-1} x, \quad dv = dx, \quad du = \frac{1}{1 + x^2} dx, \quad v = x$$

Thus,

$$\int_0^1 \tan^{-1} x\,dx = \int_0^1 u\,dv = uv\Big]_0^1 - \int_0^1 v\,du$$

$$= x\tan^{-1} x\Big]_0^1 - \int_0^1 \frac{x}{1+x^2}\,dx$$

But

$$\int_0^1 \frac{x}{1+x^2}\,dx = \frac{1}{2}\int_0^1 \frac{2x}{1+x^2}\,dx = \frac{1}{2}\ln(1+x^2)\Big]_0^1 = \frac{1}{2}\ln 2$$

so

$$\int_0^1 \tan^{-1} x\,dx = x\tan^{-1} x\Big]_0^1 - \frac{1}{2}\ln 2 = \left(\frac{\pi}{4} - 0\right) - \frac{1}{2}\ln 2 = \frac{\pi}{4} - \ln\sqrt{2}$$

□

Reduction Formulas

$$\int \sin^n x\,dx = -\frac{1}{n}\sin^{n-1} x\cos x + \frac{n-1}{n}\int \sin^{n-2} x\,dx \qquad (9)$$

$$\int \cos^n x\,dx = \frac{1}{n}\cos^{n-1} x\sin x + \frac{n-1}{n}\int \cos^{n-2} x\,dx \qquad (10)$$

To illustrate how such formulas can be obtained, let us derive (10). We begin by w
$\cos^n x$ as $\cos^{n-1} x \cdot \cos x$ and letting

$$u = \cos^{n-1} x \qquad dv = \cos x\, dx$$

$$\begin{aligned} du &= (n-1)\cos^{n-2} x(-\sin x)\, dx \qquad v = \sin x \\ &= -(n-1)\cos^{n-2} x \sin x\, dx \end{aligned}$$

so that

$$\begin{aligned} \int \cos^n x\, dx = \int \cos^{n-1} x \cos x\, dx = \int u\, dv = uv - \int v\, du \\ = \cos^{n-1} x \sin x + (n-1)\int \sin^2 x \cos^{n-2} x\, dx \\ = \cos^{n-1} x \sin x + (n-1)\int (1-\cos^2 x)\cos^{n-2} x\, dx \\ = \cos^{n-1} x \sin x + (n-1)\int \cos^{n-2} x\, dx - (n-1)\int \cos^n x\, dx \end{aligned}$$

Moving the last term on the right to the left side yields

$$n\int \cos^n x\, dx = \cos^{n-1} x \sin x + (n-1)\int \cos^{n-2} x\, dx$$

Example 8 Evaluate $\int \cos^4 x\, dx$

Solution

$$\int \cos^4 x\, dx = \tfrac{1}{4}\cos^3 x \sin x + \tfrac{3}{4}\int \cos^2 x\, dx \qquad \boxed{\text{Now apply (10) with } n = 2.}$$

$$= \tfrac{1}{4}\cos^3 x \sin x + \tfrac{3}{4}\left(\tfrac{1}{2}\cos x \sin x + \tfrac{1}{2}\int dx\right)$$

$$= \tfrac{1}{4}\cos^3 x \sin x + \tfrac{3}{8}\cos x \sin x + \tfrac{3}{8}x + C$$

$\square$

Exercise 7.2

1. $\int xe^{-2x}\,dx$

2. $\int xe^{3x}\,dx$

3. $\int x^2e^x\,dx$

4. $\int x^2e^{-2x}\,dx$

5. $\int x\sin 3x\,dx$

6. $\int x\cos 2x\,dx$

7. $\int x^2\cos x\,dx$

8. $\int x^2\sin x\,dx$

9. $\int x\ln x\,dx$

10. $\int \sqrt{x}\ln x\,dx$

11. $\int (\ln x)^2\,dx$

12. $\int \frac{\ln x}{\sqrt{x}}\,dx$

13. $\int \ln(3x-2)\,dx$

14. $\int \ln(x^2+4)\,dx$

15. $\int \sin^{-1}x\,dx$

16. $\int \cos^{-1}(2x)\,dx$

17. $\int \tan^{-1}(3x)\,dx$

18. $\int x\tan^{-1}x\,dx$

19. $\int e^x\sin x\,dx$

20. $\int e^{3x}\cos 2x\,dx$

21. $\int \sin(\ln x)\,dx$

22. $\int \cos(\ln x)\,dx$

23. $\int x\sec^2x\,dx$

24. $\int x\tan^2x\,dx$

25. $\int x^3e^{x^2}\,dx$

26. $\int \frac{xe^x}{(x+1)^2}\,dx$

27. $\int_0^2 xe^{2x}\,dx$

28. $\int_0^1 xe^{-5x}\,dx$

55. (a) Find the area of the region enclosed by $y = \ln x$, the line $x = e$, and the x-axis.

(b) Find the volume of the solid generated when the region in part (a) is revolved about the x-axis.

56. Find the area of the region between $y = x \sin x$ and $y = x$ for $0 \le x \le \pi/2$.

57. Find the volume of the solid generated when the region between $y = \sin x$ and $y = 0$ for $0 \le x \le \pi$ is revolved about the y-axis.

58. Find the volume of the solid generated when the region enclosed between $y = \cos x$ and $y = 0$ for $0 \le x \le \pi/2$ is revolved about the y-axis.

Chapter 7.3 Integrating Trigonometric Functions

Integrating Powers of Sine and Cosine

We begin by recalling two reduction formulas from the preceding section.

$$\int \sin^n x\,dx = -\frac{1}{n}\sin^{n-1} x\cos x + \frac{n-1}{n}\int \sin^{n-2} x\,dx \tag{1}$$

$$\int \cos^n x\,dx = \frac{1}{n}\cos^{n-1} x\sin x + \frac{n-1}{n}\int \cos^{n-2} x\,dx \tag{2}$$

In the case where $n = 2$, these formulas yield

$$\int \sin^2 x\,dx = -\tfrac{1}{2}\sin x\cos x + \tfrac{1}{2}\int dx = \tfrac{1}{2}x - \tfrac{1}{2}\sin x\cos x + C \tag{3}$$

$$\int \cos^2 x\,dx = \tfrac{1}{2}\cos x\sin x + \tfrac{1}{2}\int dx = \tfrac{1}{2}x + \tfrac{1}{2}\sin x\cos x + C \tag{4}$$

Alternative forms of these integration formulas can be derived from the trigonometric identities

$$\sin^2 x = \tfrac{1}{2}(1-\cos 2x) \quad \text{and} \quad \cos^2 x = \tfrac{1}{2}(1+\cos 2x) \tag{5–6}$$

which follow from the double-angle formulas

$$\cos 2x = 1 - 2\sin^2 x \quad \text{and} \quad \cos 2x = 2\cos^2 x - 1$$

These identities yield

$$\int \sin^2 x\,dx = \tfrac{1}{2}\int (1-\cos 2x)\,dx = \tfrac{1}{2}x - \tfrac{1}{4}\sin 2x + C \tag{7}$$

$$\int \cos^2 x\,dx = \tfrac{1}{2}\int (1+\cos 2x)\,dx = \tfrac{1}{2}x + \tfrac{1}{4}\sin 2x + C \tag{8}$$

In the case where $n = 3$, the reduction formulas for integrating $\sin^3 x$ and $\cos^3 x$ yield

$$\int \sin^3 x\,dx = -\frac{1}{3}\sin^2 x\cos x + \frac{2}{3}\int \sin x\,dx = -\frac{1}{3}\sin^2 x\cos x - \frac{2}{3}\cos x + C \qquad (9)$$

$$\int \cos^3 x\,dx = \frac{1}{3}\cos^2 x\sin x + \frac{2}{3}\int \cos x\,dx = \frac{1}{3}\cos^2 x\sin x + \frac{2}{3}\sin x + C \qquad (10)$$

By using the identities,

$$\sin^2 x = 1 - \cos^2 x \text{ and } \cos^2 x = 1 - \sin^2 x$$

$$\int \sin^3 x\,dx = \frac{1}{3}\cos^3 x - \cos x + C \qquad (11)$$

$$\int \cos^3 x\,dx = \sin x - \frac{1}{3}\sin^3 x + C \qquad (12)$$

H.W.

$$\int \sin^4 x\,dx = \frac{3}{8}x - \frac{1}{4}\sin 2x + \frac{1}{32}\sin 4x + C \qquad (13)$$

$$\int \cos^4 x\,dx = \frac{3}{8}x + \frac{1}{4}\sin 2x + \frac{1}{32}\sin 4x + C \qquad (14)$$

Example 1 Find the volume V of the solid that is obtained when the region under the curve $y = \sin^2 x$ over the interval $[0, \pi]$ is revolved about the x-axis (Figure 7.3.1).

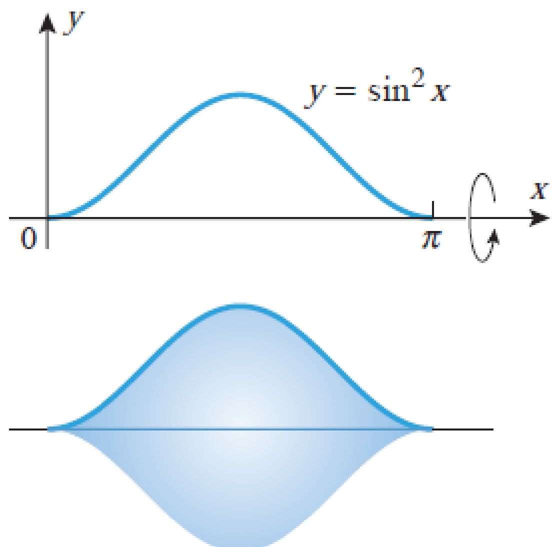


Figure 7.3.1

Using the method of disks, Formula (5) of Section 6.2, and Formula (13) above

$$V = \int_0^{\pi} \pi \sin^4 x \, dx = \pi \left[\tfrac{3}{8}x - \tfrac{1}{4}\sin 2x + \tfrac{1}{32}\sin 4x\right]_0^{\pi} = \tfrac{3}{8}\pi^2$$

Integrating Products of Sines and Cosines

If m and n are positive integers, then the integral

$$\int \sin^m x \cos^n x \, dx$$

can be evaluated by one of the three procedures stated in Table 7.3.1, depending on whether m and n are odd or even.

Table 7.3.1

INTEGRATING PRODUCTS OF SINES AND COSINES

$\int \sin^m x \cos^n x \, dx$	PROCEDURE	RELEVANT IDENTITIES
n odd	• Split off a factor of $\cos x$. • Apply the relevant identity. • Make the substitution $u = \sin x$.	$\cos^2 x = 1 - \sin^2 x$
m odd	• Split off a factor of $\sin x$. • Apply the relevant identity. • Make the substitution $u = \cos x$.	$\sin^2 x = 1 - \cos^2 x$
$\begin{cases} m \text{ even} \\ n \text{ even} \end{cases}$	• Use the relevant identities to reduce the powers on $\sin x$ and $\cos x$.	$\begin{cases} \sin^2 x = \frac{1}{2}(1 - \cos 2x) \\ \cos^2 x = \frac{1}{2}(1 + \cos 2x) \end{cases}$

Example 2 Evaluate

$$\text{(a)} \int \sin^4 x \cos^5 x \, dx \qquad \text{(b)} \int \sin^4 x \cos^4 x \, dx$$

Solution (a)

$$\begin{aligned}
\int \sin^4 x \cos^5 x \, dx &= \int \sin^4 x \cos^4 x \cos x \, dx \\
&= \int \sin^4 x (1 - \sin^2 x)^2 \cos x \, dx \\
&= \int u^4 (1 - u^2)^2 \, du \\
&= \int (u^4 - 2u^6 + u^8) \, du \\
&= \tfrac{1}{5}u^5 - \tfrac{2}{7}u^7 + \tfrac{1}{9}u^9 + C \\
&= \tfrac{1}{5}\sin^5 x - \tfrac{2}{7}\sin^7 x + \tfrac{1}{9}\sin^9 x + C
\end{aligned}$$

Solution (b)

$$\int \sin^4 x \cos^4 x\, dx = \int (\sin^2 x)^2 (\cos^2 x)^2\, dx$$

$$= \int \left(\tfrac{1}{2}[1 - \cos 2x]\right)^2 \left(\tfrac{1}{2}[1 + \cos 2x]\right)^2 dx$$

$$= \tfrac{1}{16} \int (1 - \cos^2 2x)^2\, dx$$

$$= \tfrac{1}{16} \int \sin^4 2x\, dx$$

Note that this can be obtained more directly from the original integral using the identity $\sin x \cos x = \frac{1}{2} \sin 2x$.

$$= \tfrac{1}{32} \int \sin^4 u\, du$$

$u = 2x$
$du = 2\,dx$ or $dx = \frac{1}{2}\, du$

$$= \tfrac{1}{32} \left(\tfrac{3}{8}u - \tfrac{1}{4} \sin 2u + \tfrac{1}{32} \sin 4u\right) + C$$

Formula (13)

$$= \tfrac{3}{128}x - \tfrac{1}{128} \sin 4x + \tfrac{1}{1024} \sin 8x + C$$

Integrals of the form

$$\int \sin mx \cos nx\, dx, \quad \int \sin mx \sin nx\, dx, \quad \int \cos mx \cos nx\, dx \tag{15}$$

can be found by using the trigonometric identities

$$\sin \alpha \cos \beta = \tfrac{1}{2}[\sin(\alpha - \beta) + \sin(\alpha + \beta)] \tag{16}$$

$$\sin \alpha \sin \beta = \tfrac{1}{2}[\cos(\alpha - \beta) - \cos(\alpha + \beta)] \tag{17}$$

$$\cos \alpha \cos \beta = \tfrac{1}{2}[\cos(\alpha - \beta) + \cos(\alpha + \beta)] \tag{18}$$

to express the integrand as a sum or difference of sines and cosines.

Example 3 Evaluate $\int \sin 7x \cos 3x\, dx$

Solution

$$\int \sin 7x \cos 3x\, dx = \tfrac{1}{2} \int (\sin 4x + \sin 10x)\, dx = -\tfrac{1}{8} \cos 4x - \tfrac{1}{20} \cos 10x + C$$

Integrating Powers of Tangent and Secant

$$\int \tan^n x\,dx = \frac{\tan^{n-1} x}{n-1} - \int \tan^{n-2} x\,dx \tag{19}$$

$$\int \sec^n x\,dx = \frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1}\int \sec^{n-2} x\,dx \tag{20}$$

In the case where n is odd, the exponent can be reduced to 1, leaving us with the problem of integrating $\tan x$ or $\sec x$. These integrals are given by

$$\int \tan x\,dx = \ln|\sec x| + C \tag{21}$$

$$\int \sec x\,dx = \ln|\sec x + \tan x| + C \tag{22}$$

The following basic integrals occur frequently and are worth noting:

$$\int \tan^2 x\,dx = \tan x - x + C \tag{23}$$

$$\int \sec^2 x\,dx = \tan x + C \tag{24}$$

The formulas

$$\int \tan^3 x\,dx = \tfrac{1}{2}\tan^2 x - \ln|\sec x| + C \tag{25}$$

$$\int \sec^3 x\,dx = \tfrac{1}{2}\sec x \tan x + \tfrac{1}{2}\ln|\sec x + \tan x| + C \tag{26}$$

can be deduced from (21), (22), and reduction formulas (19) and (20) as follows:

$$\int \tan^3 x\,dx = \tfrac{1}{2}\tan^2 x - \int \tan x\,dx = \tfrac{1}{2}\tan^2 x - \ln|\sec x| + C$$

$$\int \sec^3 x\,dx = \tfrac{1}{2}\sec x \tan x + \tfrac{1}{2}\int \sec x\,dx = \tfrac{1}{2}\sec x \tan x + \tfrac{1}{2}\ln|\sec x + \tan x| + C$$

Table 7.3.2

INTEGRATING PRODUCTS OF TANGENTS AND SECANTS

$\int \tan^m x \sec^n x\, dx$	PROCEDURE	RELEVANT IDENTITIES
n even	• Split off a factor of $\sec^2 x$. • Apply the relevant identity. • Make the substitution $u = \tan x$.	$\sec^2 x = \tan^2 x + 1$
m odd	• Split off a factor of $\sec x \tan x$. • Apply the relevant identity. • Make the substitution $u = \sec x$.	$\tan^2 x = \sec^2 x - 1$
$\begin{cases} m \text{ even} \\ n \text{ odd} \end{cases}$	• Use the relevant identities to reduce the integrand to powers of $\sec x$ alone. • Then use the reduction formula for powers of $\sec x$.	$\tan^2 x = \sec^2 x - 1$

Example 4

(a) $\displaystyle\int \tan^2 x \sec^4 x\, dx$ (b) $\displaystyle\int \tan^3 x \sec^3 x\, dx$ (c) $\displaystyle\int \tan^2 x \sec x\, dx$

Solution (a)

$$\begin{aligned}
\int \tan^2 x \sec^4 x\, dx &= \int \tan^2 x \sec^2 x \sec^2 x\, dx \\
&= \int \tan^2 x (\tan^2 x + 1) \sec^2 x\, dx \\
&= \int u^2(u^2 + 1)\, du \\
&= \tfrac{1}{5}u^5 + \tfrac{1}{3}u^3 + C = \tfrac{1}{5}\tan^5 x + \tfrac{1}{3}\tan^3 x + C
\end{aligned}$$

Solution (b)

$$\int \tan^3 x \sec^3 x\,dx = \int \tan^2 x \sec^2 x(\sec x \tan x)\,dx$$

$$= \int (\sec^2 x - 1)\sec^2 x(\sec x \tan x)\,dx$$

$$= \int (u^2 - 1)u^2\,du$$

$$= \tfrac{1}{5}u^5 - \tfrac{1}{3}u^3 + C = \tfrac{1}{5}\sec^5 x - \tfrac{1}{3}\sec^3 x + C$$

Solution (c)

$$\int \tan^2 x \sec x\,dx = \int (\sec^2 x - 1)\sec x\,dx$$

$$= \int \sec^3 x\,dx - \int \sec x\,dx$$

See (26) and (22)

$$= \tfrac{1}{2}\sec x \tan x + \tfrac{1}{2}\ln|\sec x + \tan x| - \ln|\sec x + \tan x| + C$$

$$= \tfrac{1}{2}\sec x \tan x - \tfrac{1}{2}\ln|\sec x + \tan x| + C$$

An Alternative Method for Integrating Powers of Sine, Cosine, Tangent, and Secant

The methods in Tables 7.3.1 and 7.3.2 can sometimes be applied if $m = 0$ or $n = 0$ to integrate positive integer powers of sine, cosine, tangent, and secant without reduction formulas. For example, instead of using the reduction formula to integrate $\sin^3 x$, we can apply the second procedure in Table 7.3.1:

$$\begin{aligned}\int \sin^3 x\,dx &= \int (\sin^2 x)\sin x\,dx \\ &= \int (1-\cos^2 x)\sin x\,dx \qquad \boxed{\begin{aligned} u &= \cos x \\ du &= -\sin x\,dx\end{aligned}} \\ &= -\int (1-u^2)\,du \\ &= \tfrac{1}{3}u^3 - u + C = \tfrac{1}{3}\cos^3 x - \cos x + C\end{aligned}$$

Exercise Set 7.3 Evaluate the integral

1. $\int \cos^3 x \sin x\,dx$

2. $\int \sin^5 3x \cos 3x\,dx$

3. $\int \sin^2 5\theta\,d\theta$

4. $\int \cos^2 3x\,dx$

5. $\int \sin^3 a\theta\,d\theta$

6. $\int \cos^3 at\,dt$

7. $\int \sin ax \cos ax\,dx$

8. $\int \sin^3 x \cos^3 x\,dx$

9. $\int \sin^2 t \cos^3 t\,dt$

10. $\int \sin^3 x \cos^2 x\,dx$

11. $\int \sin^2 x \cos^2 x \, dx$

12. $\int \sin^2 x \cos^4 x \, dx$

13. $\int \sin 2x \cos 3x \, dx$

14. $\int \sin 3\theta \cos 2\theta \, d\theta$

15. $\int \sin x \cos(x/2) \, dx$

16. $\int \cos^{1/3} x \sin x \, dx$

17. $\int_0^{\pi/2} \cos^3 x \, dx$

18. $\int_0^{\pi/2} \sin^2 \frac{x}{2} \cos^2 \frac{x}{2}$

19. $\int_0^{\pi/3} \sin^4 3x \cos^3 3x \, dx$

20. $\int_{-\pi}^{\pi} \cos^2 5\theta \, d\theta$

21. $\int_0^{\pi/6} \sin 4x \cos 2x \, dx$

22. $\int_0^{2\pi} \sin^2 kx \, dx$

23. $\int \sec^2(2x - 1) \, dx$

24. $\int \tan 5x \, dx$

25. $\int e^{-x} \tan(e^{-x}) \, dx$

26. $\int \cot 3x \, dx$

27. $\int \sec 4x \, dx$

28. $\int \frac{\sec(\sqrt{x})}{\sqrt{x}} \, dx$

29. $\int \tan^2 x \sec^2 x \, dx$

30. $\int \tan^5 x \sec^4 x \, dx$

Chapter 7.4 Trigonometric Substitutions

To start, we will be concerned with integrals that contain expressions of the form

$$\sqrt{a^2-x^2}, \quad \sqrt{x^2+a^2}, \quad \sqrt{x^2-a^2}$$

in which a is a positive constant. The basic idea for evaluating such integrals is to make a substitution for x that will eliminate the radical. For example, to eliminate the radical in the expression $\sqrt{a^2-x^2}$, we can make the substitution

$$x = a\sin\theta, \quad -\pi/2 \le \theta \le \pi/2 \tag{1}$$

which yields

$$\sqrt{a^2-x^2} = \sqrt{a^2-a^2\sin^2\theta} = \sqrt{a^2(1-\sin^2\theta)}$$

$$= a\sqrt{\cos^2\theta} = a|\cos\theta| = a\cos\theta \qquad \boxed{\cos\theta \ge 0 \text{ since } -\pi/2 \le \theta \le \pi/2}$$

Example 1 Evaluate

$$\int \frac{dx}{x^2\sqrt{4-x^2}}$$

Solution

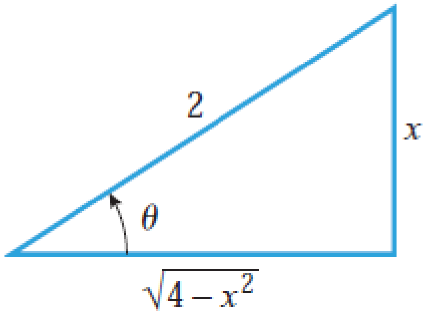
To eliminate the radical we make the substitution

$$x = 2\sin\theta, \quad dx = 2\cos\theta\, d\theta$$

$$\int \frac{dx}{x^2\sqrt{4-x^2}} = \int \frac{2\cos\theta\, d\theta}{(2\sin\theta)^2\sqrt{4-4\sin^2\theta}}$$

$$= \int \frac{2\cos\theta\, d\theta}{(2\sin\theta)^2(2\cos\theta)} = \frac{1}{4}\int \frac{d\theta}{\sin^2\theta}$$

$$= \frac{1}{4}\int \csc^2\theta\, d\theta = -\frac{1}{4}\cot\theta + C$$



$\boxed{x = 2\sin\theta}$

$$\sin\theta = x/2$$

$$\cot\theta = \frac{\sqrt{4-x^2}}{x}$$

Example 2 Evaluate

$$\int_1^{\sqrt{2}} \frac{dx}{x^2\sqrt{4-x^2}}$$

Solution

Method 1.

Using the result from Example 1 with the x-limits of integration yields

$$\int_1^{\sqrt{2}} \frac{dx}{x^2\sqrt{4-x^2}} = -\frac{1}{4}\left[\frac{\sqrt{4-x^2}}{x}\right]_1^{\sqrt{2}} = -\frac{1}{4}\left[1-\sqrt{3}\right] = \frac{\sqrt{3}-1}{4}$$

Method 2.

The substitution $x = 2\sin\theta$ can be expressed as $x/2 = \sin\theta$ or $\theta = \sin^{-1}(x/2)$, so the θ-limits that correspond to $x = 1$ and $x = \sqrt{2}$ are

$$x = 1: \quad \theta = \sin^{-1}(1/2) = \pi/6$$

$$x = \sqrt{2}: \quad \theta = \sin^{-1}(\sqrt{2}/2) = \pi/4$$

Thus, from (2) in Example 1 we obtain

$$\int_1^{\sqrt{2}} \frac{dx}{x^2\sqrt{4-x^2}} = \frac{1}{4}\int_{\pi/6}^{\pi/4} \csc^2\theta \, d\theta$$

Convert x-limits to θ-limits.

$$= -\frac{1}{4}\left[\cot\theta\right]_{\pi/6}^{\pi/4} = -\frac{1}{4}\left[1-\sqrt{3}\right] = \frac{\sqrt{3}-1}{4}$$

Example 3 (H.W)

Find the area of the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Table 7.4.1

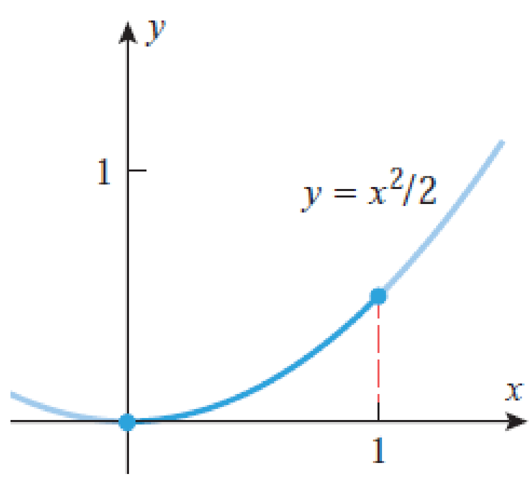
TRIGONOMETRIC SUBSTITUTIONS

EXPRESSION IN THE INTEGRAND	SUBSTITUTION	RESTRICTION ON θ	SIMPLIFICATION
$\sqrt{a^2 - x^2}$	$x = a\sin\theta$	$-\pi/2 \le \theta \le \pi/2$	$a^2 - x^2 = a^2 - a^2\sin^2\theta = a^2\cos^2\theta$
$\sqrt{a^2 + x^2}$	$x = a\tan\theta$	$-\pi/2 < \theta < \pi/2$	$a^2 + x^2 = a^2 + a^2\tan^2\theta = a^2\sec^2\theta$
$\sqrt{x^2 - a^2}$	$x = a\sec\theta$	$\begin{cases} 0 \le \theta < \pi/2 & (\text{if } x \ge a) \\ \pi/2 < \theta \le \pi & (\text{if } x \le -a) \end{cases}$	$x^2 - a^2 = a^2\sec^2\theta - a^2 = a^2\tan^2\theta$

Example 4

Find the arc length of the curve $y = x^2/2$ from $x = 0$ to $x = 1$

Solution



From Formula (4) of Section 6.4 the arc length L of the curve is

$$L = \int_0^1 \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx = \int_0^1 \sqrt{1 + x^2}\, dx$$

The integrand involves a radical of the form $\sqrt{a^2 + x^2}$ with $a = 1$, so from Table 7.4.1 we make the substitution

$$x = \tan\theta, \quad -\pi/2 < \theta < \pi/2$$

$$\frac{dx}{d\theta} = \sec^2\theta \quad \text{or} \quad dx = \sec^2\theta\, d\theta$$

Since this substitution can be expressed as $\theta = \tan^{-1} x$, the θ-limits of integration that correspond to the x-limits, $x = 0$ and $x = 1$, are

$$x = 0: \quad \theta = \tan^{-1} 0 = 0$$

$$x = 1: \quad \theta = \tan^{-1} 1 = \pi/4$$

Thus,

$$\begin{aligned}
L = \int_0^1 \sqrt{1 + x^2}\, dx &= \int_0^{\pi/4} \sqrt{1 + \tan^2\theta}\, \sec^2\theta\, d\theta \\
&= \int_0^{\pi/4} \sqrt{\sec^2\theta}\, \sec^2\theta\, d\theta \qquad \boxed{1 + \tan^2\theta = \sec^2\theta} \\
&= \int_0^{\pi/4} |\sec\theta| \sec^2\theta\, d\theta \\
&= \int_0^{\pi/4} \sec^3\theta\, d\theta \qquad \boxed{\sec\theta > 0 \text{ since } -\pi/2 < \theta < \pi/2} \\
&= \left[\tfrac{1}{2}\sec\theta\tan\theta + \tfrac{1}{2}\ln|\sec\theta + \tan\theta|\right]_0^{\pi/4} \qquad \boxed{\text{Formula (26) of Section 7.3}} \\
&= \tfrac{1}{2}\left[\sqrt{2} + \ln(\sqrt{2} + 1)\right] \approx 1.148
\end{aligned}$$

Example 5

$$\text{Evaluate} \int \frac{\sqrt{x^2-25}}{x}\,dx, \text{ assuming that } x \geq 5.$$

Solution

Table 7.4.1 we make the substitution

$$x = 5\sec\theta, \quad 0 \leq \theta < \pi/2$$

$$\frac{dx}{d\theta} = 5\sec\theta\tan\theta \quad \text{or} \quad dx = 5\sec\theta\tan\theta\,d\theta$$

Thus,

$$\int \frac{\sqrt{x^2-25}}{x}\,dx = \int \frac{\sqrt{25\sec^2\theta-25}}{5\sec\theta}(5\sec\theta\tan\theta)\,d\theta$$

$$= \int \frac{5|\tan\theta|}{5\sec\theta}(5\sec\theta\tan\theta)\,d\theta$$

$$= 5\int \tan^2\theta\,d\theta \qquad \boxed{\tan\theta \geq 0 \text{ since } 0 \leq \theta < \pi/2}$$

$$= 5\int (\sec^2\theta - 1)\,d\theta = 5\tan\theta - 5\theta + C$$

To express the solution in terms of x, we will represent the substitution $x = 5\sec\theta$ geometrically by the triangle in Figure 7.4.5, from which we obtain

$$\tan\theta = \frac{\sqrt{x^2-25}}{5}$$

From this and the fact that the substitution can be expressed as $\theta = \sec^{-1}(x/5)$, we obtain

$$\int \frac{\sqrt{x^2-25}}{x}\,dx = \sqrt{x^2-25} - 5\sec^{-1}\left(\frac{x}{5}\right) + C \blacktriangleleft$$

Example 5

$$\text{Evaluate} \int \frac{x}{x^2-4x+8}\,dx$$

Solution

$$x^2-4x+8=(x^2-4x+4)+8-4=(x-2)^2+4$$

Thus, the substitution

$$u=x-2, \quad du=dx$$

yields

$$\begin{aligned}
\int \frac{x}{x^2-4x+8}\,dx &= \int \frac{x}{(x-2)^2+4}\,dx = \int \frac{u+2}{u^2+4}\,du \\
&= \int \frac{u}{u^2+4}\,du + 2\int \frac{du}{u^2+4} \\
&= \frac{1}{2}\int \frac{2u}{u^2+4}\,du + 2\int \frac{du}{u^2+4} \\
&= \frac{1}{2}\ln(u^2+4) + 2\left(\frac{1}{2}\right)\tan^{-1}\frac{u}{2} + C \\
&= \frac{1}{2}\ln[(x-2)^2+4] + \tan^{-1}\left(\frac{x-2}{2}\right) + C
\end{aligned}$$

Exercise Set 7.4 Evaluate the integral

1. $\int \sqrt{4-x^2}\,dx$

2. $\int \sqrt{1-4x^2}\,dx$

3. $\int \frac{x^2}{\sqrt{16-x^2}}\,dx$

4. $\int \frac{dx}{x^2\sqrt{9-x^2}}$

5. $\int \frac{dx}{(4+x^2)^2}$

6. $\int \frac{x^2}{\sqrt{5+x^2}}\,dx$

7. $\int \frac{\sqrt{x^2-9}}{x}\,dx$

8. $\int \frac{dx}{x^2\sqrt{x^2-16}}$

9. $\int \frac{3x^3}{\sqrt{1-x^2}}\,dx$

10. $\int x^3\sqrt{5-x^2}\,dx$

11. $\int \frac{dx}{x^2\sqrt{9x^2-4}}$

12. $\int \frac{\sqrt{1+t^2}}{t}\,dt$

13. $\int \frac{dx}{(1-x^2)^{3/2}}$

14. $\int \frac{dx}{x^2\sqrt{x^2+25}}$

15. $\int \frac{dx}{\sqrt{x^2-9}}$

16. $\int \frac{dx}{1+2x^2+x^4}$

17. $\int \frac{dx}{(4x^2-9)^{3/2}}$

18. $\int \frac{3x^3}{\sqrt{x^2-25}}\,dx$

19. $\int e^x\sqrt{1-e^{2x}}\,dx$

20. $\int \frac{\cos\theta}{\sqrt{2-\sin^2\theta}}\,d\theta$

33. Find the arc length of the curve $y = \ln x$ from $x = 1$ to $x = 2$.

34. Find the arc length of the curve $y = x^2$ from $x = 0$ to $x = 1$.

36. Find the volume of the solid generated when the region enclosed by $x = y(1 - y^2)^{1/4}$, $y = 0$, $y = 1$, and $x = 0$ is revolved about the y-axis.

(Assignment - 33, 34)!

Evaluate the integral

37. $\displaystyle\int \frac{dx}{x^2 - 4x + 5}$

38. $\displaystyle\int \frac{dx}{\sqrt{2x - x^2}}$

39. $\displaystyle\int \frac{dx}{\sqrt{3 + 2x - x^2}}$

40. $\displaystyle\int \frac{dx}{16x^2 + 16x + 5}$

41. $\displaystyle\int \frac{dx}{\sqrt{x^2 - 6x + 10}}$

42. $\displaystyle\int \frac{x}{x^2 + 2x + 2}\, dx$

7.5 Integrating Rational Functions by Partial Fractions

Partial Fractions

$$\frac{5x - 10}{(x-4)(x+1)} = \frac{A}{x-4} + \frac{B}{x+1}$$

$$5x - 10 = A(x + 1) + B(x - 4)$$

Equating the coefficient of x and constant terms and then solving we obtain $A = 2\,, B = 3$.

$$\frac{5x - 10}{(x-4)(x+1)} = \frac{2}{x-4} + \frac{3}{x+1}$$

$$\int \frac{5x-10}{x^2-3x-4}\,dx = \int \frac{2}{x-4}\,dx + \int \frac{3}{x+1}\,dx$$

$$= 2\ln|x-4| + 3\ln|x+1| + C$$

Example 1

$$\text{Evaluate } \int \frac{dx}{x^2+x-2}.$$

Solution

$$\frac{1}{x^2+x-2} = \frac{1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$$

$$1 = A(x+2) + B(x-1)$$

$$\Rightarrow \quad A = \tfrac{1}{3}; \text{ and } B = -\tfrac{1}{3}.$$

$$\Rightarrow \quad \frac{1}{(x-1)(x+2)} = \frac{\frac{1}{3}}{x-1} + \frac{-\frac{1}{3}}{x+2}$$

The integration can now be completed as follows:

$$\int \frac{dx}{(x-1)(x+2)} = \frac{1}{3}\int \frac{dx}{x-1} - \frac{1}{3}\int \frac{dx}{x+2}$$

$$= \frac{1}{3}\ln|x-1| - \frac{1}{3}\ln|x+2| + C = \frac{1}{3}\ln\left|\frac{x-1}{x+2}\right| + C$$

$\square$

Example 2

$$\text{Evaluate } \int \frac{2x+4}{x^3-2x^2}\,dx.$$

Solution

$$\frac{2x+4}{x^2(x-2)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-2}$$

$$\Rightarrow 2x+4 = Ax(x-2) + B(x-2) + Cx^2$$

$$\Rightarrow 2x+4 = (A+C)x^2 + (-2A+B)x - 2B$$

$$\Rightarrow A=-2,\ B=-2, \text{ and } C=2$$

$$\Rightarrow \frac{2x+4}{x^2(x-2)} = \frac{-2}{x} + \frac{-2}{x^2} + \frac{2}{x-2}$$

$$\int \frac{2x+4}{x^2(x-2)}\,dx = -2\int \frac{dx}{x} - 2\int \frac{dx}{x^2} + 2\int \frac{dx}{x-2}$$

$$= -2\ln|x| + \frac{2}{x} + 2\ln|x-2| + C = 2\ln\left|\frac{x-2}{x}\right| + \frac{2}{x} + C$$

$\square$

Example 3

$$\text{Evaluate } \int \frac{x^2+x-2}{3x^3-x^2+3x-1}\,dx.$$

$$\frac{x^2+x-2}{(3x-1)(x^2+1)} = \frac{A}{3x-1} + \frac{Bx+C}{x^2+1}$$

$$x^2+x-2 = A(x^2+1) + (Bx+C)(3x-1)$$

$$x^2+x-2 = (A+3B)x^2 + (-B+3C)x + (A-C)$$

$$\begin{aligned} A + 3B \quad\quad &= 1 \\ -\ B + 3C &= 1 \\ A \quad\quad -\ C &= -2 \end{aligned} \quad \Rightarrow A = -\frac{7}{5},\quad B = \frac{4}{5},\quad C = \frac{3}{5}$$

$$\frac{x^2+x-2}{(3x-1)(x^2+1)} = \frac{-\frac{7}{5}}{3x-1} + \frac{\frac{4}{5}x+\frac{3}{5}}{x^2+1}$$

$$\int \frac{x^2+x-2}{(3x-1)(x^2+1)}\,dx = -\frac{7}{5}\int \frac{dx}{3x-1} + \frac{4}{5}\int \frac{x}{x^2+1}\,dx + \frac{3}{5}\int \frac{dx}{x^2+1}$$

$$= -\frac{7}{15}\ln|3x-1| + \frac{2}{5}\ln(x^2+1) + \frac{3}{5}\tan^{-1}x + C$$

$\square$

Integrating Improper Rational Functions

Example 5

$$\text{Evaluate } \int \frac{3x^4+3x^3-5x^2+x-1}{x^2+x-2}\,dx$$

$$\begin{array}{r|l}
 & \quad\quad\quad\quad\;\; 3x^2 \quad\quad + 1 \\
\cline{2-2}
x^2+x-2 & 3x^4+3x^3-5x^2+x-1 \\
 & \underline{3x^4+3x^3-6x^2} \quad\quad\quad \\
 & \quad\quad\quad\quad\;\; x^2+x-1 \\
 & \quad\quad\quad\quad\;\; \underline{x^2+x-2} \\
 & \quad\quad\quad\quad\quad\quad\quad\;\; 1
\end{array}$$

$$\frac{3x^4+3x^3-5x^2+x-1}{x^2+x-2} = (3x^2+1) + \frac{1}{x^2+x-2}$$

$$\int \frac{3x^4+3x^3-5x^2+x-1}{x^2+x-2}\,dx = \int (3x^2+1)\,dx + \int \frac{dx}{x^2+x-2}$$

$$\int \frac{3x^4+3x^3-5x^2+x-1}{x^2+x-2}\,dx = x^3+x+\frac{1}{3}\ln\left|\frac{x-1}{x+2}\right| + C$$

$\square$

Exercise Set 7.5 Evaluate the integral

9. $\displaystyle\int \frac{dx}{x^2 - 3x - 4}$

10. $\displaystyle\int \frac{dx}{x^2 - 6x - 7}$

11. $\displaystyle\int \frac{11x + 17}{2x^2 + 7x - 4}\,dx$

12. $\displaystyle\int \frac{5x - 5}{3x^2 - 8x - 3}\,dx$

13. $\displaystyle\int \frac{2x^2 - 9x - 9}{x^3 - 9x}\,dx$

14. $\displaystyle\int \frac{dx}{x(x^2 - 1)}$

15. $\displaystyle\int \frac{x^2 - 8}{x + 3}\,dx$

16. $\displaystyle\int \frac{x^2 + 1}{x - 1}\,dx$

17. $\displaystyle\int \frac{3x^2 - 10}{x^2 - 4x + 4}\,dx$

18. $\displaystyle\int \frac{x^2}{x^2 - 3x + 2}\,dx$

19. $\displaystyle\int \frac{2x - 3}{x^2 - 3x - 10}\,dx$

20. $\displaystyle\int \frac{3x + 1}{3x^2 + 2x - 1}\,dx$

21. $\displaystyle\int \frac{x^5 + x^2 + 2}{x^3 - x}\,dx$

22. $\displaystyle\int \frac{x^5 - 4x^3 + 1}{x^3 - 4x}\,dx$

23. $\displaystyle\int \frac{2x^2 + 3}{x(x - 1)^2}\,dx$

24. $\displaystyle\int \frac{3x^2 - x + 1}{x^3 - x^2}\,dx$